

Adaptive Prompt Generation Using ChatGPT as Cognitive Scaffolding for Improving Spanish as a Foreign Language (ELE) Writing

Dexin Kong*

Xi'an Fanyi University

Xi'an, Shaanxi, China

miranda-kong@qq.com

Abstract

This study examines the use of generative artificial intelligence, specifically ChatGPT-3.5, as a pedagogical mediator to enhance writing in Spanish as a foreign language (ELE). A hierarchical five-level prompt system was implemented, manually integrated into the model's interface, simulating a conditional flow of progressive feedback. The architecture, grounded in cognitive scaffolding, was applied through controlled interactions with the Transformer model without additional training. The intervention, evaluated through a mixed quasi-experimental design ($n=60$), included automatic error analysis, codification of metacognitive strategies, and visualizations of textual behavior. The results show that the experimental group significantly reduced linguistic errors and activated more reflective processes than the control group. It is concluded that a technically structured system of hierarchical prompts can function as an adaptive computational tutoring system in AI-mediated educational environments.

CCS Concepts

• Applied computing; • Education; • Computer-assisted instruction;

Keywords

ChatGPT-3.5, Transformer architecture, multilevel prompt structures, automated feedback system, automatic linguistic error analysis, natural language processing (NLP)

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1 Introduction

Writing in the acquisition of foreign languages fulfills a dual role: as a productive skill and as a tool for metalinguistic reflection. According to Swain's Output Hypothesis, producing written text activates

deep cognitive processes—grammatical, lexical, and discursive—that receptive input alone does not stimulate. In a globalized context, writing competence is increasingly essential in both academic and professional settings. However, the teaching of writing in Spanish as a Foreign Language (ELE) continues to face persistent limitations. While teacher feedback is valuable, it often arrives with delays [1], and automated tools like Grammarly, though fast, tend to promote passive correction without fostering understanding of the error [2]. Moreover, there is a notable lack of technological resources tailored to ELE, in contrast to the abundance of solutions available for English.

In this context, Artificial Intelligence (AI) emerges as a strategic pedagogical resource. UNESCO (2021) highlights its potential to develop key 21st-century competencies, provided it is integrated for educational purposes. ChatGPT, through a system of scaffolded prompts, offers three main advantages: (1) it promotes learner autonomy by means of progressive questioning, (2) it adjusts the level of support to the user's linguistic profile, and (3) it delivers immediate feedback. Its architecture is based on large language models (LLMs) trained with the Transformer framework [3], where prompts act as initial instructions that shape the generated response. This study explores how the intentional design of prompts can function as cognitive scaffolding in ELE writing, drawing on Vygotsky's theory of the Zone of Proximal Development (1978). Through a controlled experiment, the effectiveness of this system is evaluated against traditional methods, considering both error reduction and the activation of metacognitive strategies. The findings aim to offer a replicable model for the ethical and pedagogical integration of AI in contexts where Spanish is the target language.

2 THEORETICAL FRAMEWORK

2.1 Theoretical Foundations

The integration of ChatGPT into ELE writing instruction is supported by two contemporary theoretical pillars. On one hand, Vygotsky's (1978) Zone of Proximal Development (ZPD) remains applicable in AI-assisted mediation contexts. Flores Vivar and García Peñalvo emphasize that intelligent learning environments must offer "micro levels of progressive mediation to facilitate cognitive internalization" [4], a principle that justifies the use of scaffolded prompts. On the other hand, error analysis theory has evolved toward reflective approaches. Blázquez-Carretero and Woore demonstrate that corrective tools that break down rules into micro-steps enhance metalinguistic awareness, unlike those that simply provide solutions [5]. Both frameworks support the need for reflective and tiered feedback in AI-based educational systems.

*Corresponding author



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Table 1: Five-Level Scaffolded Feedback System Based on Cognitive Scaffolding Principles

Level	Type of Prompt	Example for Verb Tense Error
L1	Error Localization	Check if the verb tense is used correctly here.
L2	Tentative Reformulation	Could you rephrase this part while keeping the meaning?
L3	Grammar Rule Reminder	Recall the difference between preterite and imperfect.
L4	Contextual Contrast	Does this tense match the previous sentence?
L5	Reflective Question	Why did you use present tense instead of past tense here?

2.2 Limitations of Current Tools

Multiple studies reveal significant deficiencies in digital resources for writing in ELE. Blázquez-Carretero notes that conventional grammar checkers tend to promote mechanical correction over actual grammatical learning [5]. A report by the Instituto Cervantes (2022) indicates that only 28% of commercial tools correctly handle language-specific phenomena such as the use of the subjunctive or the *ser/estar* distinction, compared to an estimated 90% coverage in tools developed for English [6]. Furthermore, a review by Yang Lu et al. reveals that 85% of research on ChatGPT in second language (L2) contexts focuses exclusively on English, failing to address specific issues in ELE such as negative transfer [7].

2.3 Rationale for the Prompt-Based Intervention

Considering this scenario, the design of scaffolded prompts emerges as an effective pedagogical alternative. The Production-Oriented Approach (POA) posits that forced output activates grammatical awareness and facilitates internalization of linguistic knowledge. In the context of ELE, instructional sequences that guide learners from error identification to autonomous revision promote progressive and independent learning. Recent research confirms that using ChatGPT as cognitive scaffolding enhances both grammatical accuracy and the use of metacognitive strategies, outperforming the effects of direct correction [8–10].

3 METHODOLOGY

3.1 Research Design

This study followed a quasi-experimental mixed-method design (quantitative and qualitative), focused on comparing two feedback conditions during academic writing tasks in ELE. Two parallel groups were formed: the control group received direct corrections generated by ChatGPT (free version 3.5), while the experimental group interacted with a five-level scaffolded feedback system based on principles of cognitive scaffolding (see Table 1). The primary dependent variable was the number of linguistic errors per 100 words. As a secondary variable, the activation of metacognitive strategies was analyzed based on the types of textual modifications introduced by students during their revisions.

3.2 Participants and Sample

The study involved 60 university students with a B1 level of Spanish as a Foreign Language, according to the CEFR. Participants were recruited through convenience sampling at an Asian university and randomly assigned to two equivalent groups ($n = 30$). Prior to

the intervention, all participants completed a diagnostic writing task and a grammar test, whose results confirmed initial statistical homogeneity ($t(58) = 0.74, p = .46$). The average age was 21.4 years ($SD = 1.9$), and all students had received at least three semesters of formal instruction in Spanish.

3.3 Instruments

Three main instruments were used in the study. First, ChatGPT 3.5 (free web-based interface) served as an interactive writing assistant, complemented by Google Docs in version history mode to document revisions. The writing tasks were adapted from the official DELE B1 exam and included: a complaint letter (150–180 words), a biographical narrative, and an argumentative opinion on healthy habits. The prompts followed CEFR guidelines for level B1, with minor adjustments to facilitate interaction with the AI system. To assess the metacognitive dimension, a validated learning strategies questionnaire originally designed for digital environments was applied. The instrument consisted of 18 Likert-scale items ($\alpha = .89$ in its original validation) and was linguistically adapted to Peninsular Spanish after being piloted with 15 students with profiles like the experimental group.

3.4 Technical Design of the Prompt System

The scaffolded feedback system was structured into five hierarchical levels (L1–L5), aligned with a gradual progression of cognitive complexity. Each level represented a distinct type of intervention, ranging from explicit error localization to student-led metalinguistic self-regulation. This sequence was implemented manually via the web interface of ChatGPT 3.5, using prompts written in natural language. Table 2 summarizes the five levels and their specific roles within the textual revision process:

Progression to the next level occurred only if the student failed to correct the error after receiving the previous prompt. This logic is illustrated through a conditional flow diagram (see Figure 1).

To more precisely illustrate how each level provides a distinct form of cognitive scaffolding, Table 3 presents concrete examples of the prompts used at the five hierarchical levels (higher levels are only triggered if the learner does not self-correct after the preceding prompt). These examples allow readers to grasp at a glance both the wording of each prompt and the specific cognitive function it activates during the text-revision process.

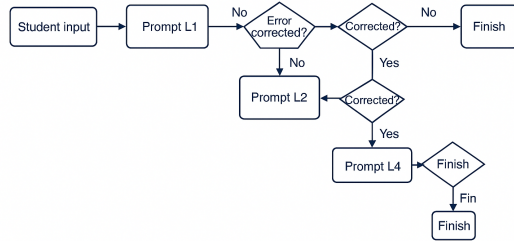
From a technical standpoint, the implementation did not require additional programming. The free version of ChatGPT-3.5 was used, with the following parameters: default temperature (1.0), a token limit of approximately 4096, and input provided as unstructured

Table 2: Hierarchical Levels of the Prompt-Based Feedback System

Level	Type of support	Prompt example
L1	Direct error localization	“Atención: error de concordancia en esta frase.”
L2	Guided rewriting	“Reescribe esta frase respetando la concordancia.”
L3	Grammar rule reminder	“¿Recuerdas cómo concuerda un adjetivo con un sustantivo?”
L4	Guided induction	“¿Cuál sería una regla aplicable en este contexto?”
L5	Open metacognitive question	“¿Por qué elegiste este tiempo verbal en esta frase?”

Table 3: Examples of Hierarchical Prompts and their Cognitive Scaffolding Functions

Level	Prompt Text (in Spanish)	Cognitive Scaffolding Function
L1	«Identifica el error de concordancia de género y número en esta frase.»	Explicit error localization
L2	¿Qué sustantivo debería estar en plural?	Lexical comparison activation
L3	«Recuerda la diferencia entre pretérito imperfecto y pretérito indefinido. ¿Cuál elegirías aquí?»	Grammatical-rules anchoring
L4	¿Cómo cambiaría el sentido si sustituyes el pretérito indefinido por el pretérito perfecto?	Contextual inference
L5	¿Por qué crees que este tiempo verbal refleja mejor la intención del autor?	Metacognitive self-regulation


Figure 1: Flowchart of the Scaffolded Prompt Feedback System

plain text. Prompts were generated manually without fine-tuning or task-specific training.

To evaluate the impact of this progressive feedback system, three key quantitative metrics were defined:

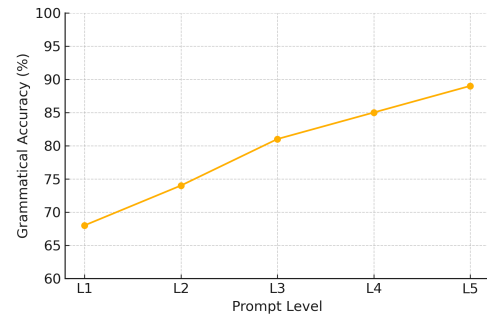
1. Overall error rate: calculated as the number of linguistic errors per 100 words written, according to formula (1), where E_t represents total errors and N_p the number of words.

$$\text{Error rate (\%)} = \left(\frac{E_t}{N_p} \times 100 \right) \quad (1)$$

2. Error distribution by type: errors were classified according to the linguistic category affected, as shown in formula (2).

$$E_{type} \in \{\text{morphosyntax, lexico, orthography}\} \quad (2)$$

3. Cognitive strategy coding: based on the modifications made by students, distinguishing between mechanical correction and


Figure 2: Relationship Between Prompt Level and Grammatical Accuracy

metalinguistic reflection (see formula 3).

$$E = \begin{bmatrix} 1 & \text{Direct} & \text{substitution} \\ 2 & \text{Contextual} & \text{restructuring} \\ 3 & \text{Reflective} & \text{inference} \end{bmatrix} \quad (3)$$

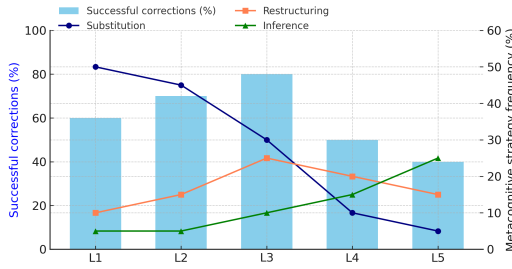
In addition, representative interactions were documented for each level (see Table 4), allowing observation of how cognitive demands increase as the intervention progresses.

Finally, two visualizations were created (see Figures 2 and 3) to support the technical analysis of the system:

These figures show that levels L2 and L3 are most effective for immediate correction, whereas L4 and L5—though less frequent—tend to promote deeper metacognitive inference. Overall, this technical design enables the articulation of scaffolded feedback within an AI-driven conversational environment, allowing for controlled cognitive variables in the revision process. Its modular structure also facilitates replication in other pedagogical settings with limited technological resources.

Table 4: Sample Interactions by Prompt Level

Level	Prompt input	ChatGPT response
L1	“Review the sentence: ‘La chica alto llegó.’”	“There’s a gender error in ‘alto’. Can you fix it?”
L2	“Correct: ‘La chica alto llegó.’”	“La chica alta llegó.”
L3	“How should an adjective agree with the noun?”	“It must agree in gender and number: ‘alta’ with ‘chica.’”
L4	“What rule explains the agreement here?”	“What rule explains the agreement here?”
L5	“Why is ‘alta’ more appropriate than ‘alto’?”	“Because ‘chica’ is feminine, so it must agree.”

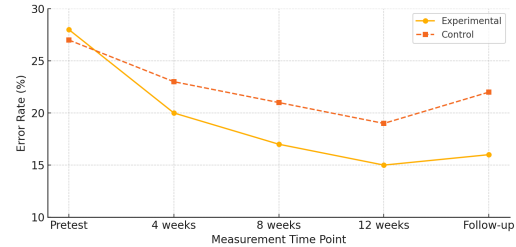
**Figure 3: Correction Effectiveness and Cognitive Activation by Prompt Level**

3.5 Procedure

Each feedback level was operationalized through structured activating prompts, which were manually entered into the ChatGPT interface. For instance, Level L3 included inputs such as: “Recall the difference between the preterite and the imperfect. Which would be more appropriate here?” whereas Level L5 posed open-ended questions like: “Why do you think this verb tense is more suitable for this narrative context?” Participants copied their text into the chat and responded according to the feedback level they received before proceeding to the next. The conditional logic of this stepped loop interaction is summarized in Figure 1 within the experimental group, if an error remained uncorrected after a first attempt, the system advanced to the next level until either L5 was reached or self-correction occurred. The study spanned twelve weeks, comprising one written-practice session per week and monthly evaluations (intermediate posttest and follow-up). During the initial session, ethical guidelines for AI use and the basic functionality of ChatGPT were presented, followed by a diagnostic task (pretest). Throughout the intervention phase (three weekly sessions), participants drafted texts, engaged with the feedback system according to their assigned group, and completed a final review. At the conclusion of the twelve weeks, a retention posttest with a different prompt was administered to assess consolidation of learning.

3.6 Data Analysis

Two complementary analyses were conducted. For the quantitative data, error indices per task (total number of errors / total words $\times 100$) were computed at three time points: pretest, intermediate posttest, and retention posttest. These values were compared using an analysis of covariance (ANCOVA), with the initial pretest score entered as a covariate to control for baseline variability. A significance threshold of $p < .05$ was applied. Additionally, a heat

**Figure 4: Evolution of Error Rates**

map was generated to visualize the frequency of activation of each prompt level over the 12-week intervention. For the qualitative data, a thematic content analysis of the revision processes was performed, categorizing strategies into three types: directed substitution, contextual restructuring, and inferential review. This analysis was triangulated with responses from a metacognitive-strategy questionnaire and two semi-structured interviews conducted with representative students from each group.

4 RESULTS

The following section presents the results organized according to the study’s specific objectives. The first objective was to analyze the impact of progressive suggestions versus direct correction on the improvement of grammatical accuracy in texts written by B1-level Spanish learners. To this end, we conducted a descriptive and comparative statistical analysis of error rates in written productions at five time points: pretest, 4-week posttest, 8-week posttest, 12-week posttest, and a follow-up assessment two weeks after the intervention. Table 5 reports the mean percentages of lexical, morphosyntactic, and orthographic errors identified by external raters at each phase.

In the experimental group, the error rate decreased from 28 % at pretest to 20 % at the 4-week posttest, 17 % at the 8-week posttest, and 15 % at the 12-week posttest, with a slight rebound to 16 % at follow-up. In contrast, the control group exhibited a more modest reduction (27 % $\rightarrow$ 23 % $\rightarrow$ 21 % $\rightarrow$ 19 %), followed by an increase to 22 % at follow-up. These findings demonstrate greater learning retention in the scaffolding-assisted group, as confirmed by ANCOVA ($F(1,118) = 9.12$; $p = 0.003$). (See Figure 4)

The second objective focused on examining the use of metacognitive strategies based on the type of prompt received. To this end, a frequency analysis was conducted using the version history in Google Docs. Level L3 prompts were the most frequently used

Table 5: Comparison of error rates (%)

Group	Pretest	4 weeks	8 weeks	12 weeks	Follow-up
Experimental	28	20	17	15	16
Control	27	23	21	19	22

(45%), followed by levels L1 (30%) and L2 (15%), while L4 and L5 were used less frequently, reflecting their more advanced nature.

At the qualitative level, two main revision strategy patterns emerged: a mechanical approach involving literal substitution, predominant in the control group, and a reflective approach based on contextual inference, more common in the experimental group. A representative case involved a student who, after receiving the L5 prompt “Why is the preterite used in this context instead of the present perfect?”, independently consulted a grammar manual and revised three similar structures in her text. This response suggests a significant activation of metalinguistic awareness, absent in the group that received direct correction. Taken together, the findings support the hypothesis that scaffolded feedback not only improves immediate grammatical accuracy but also promotes deeper and more autonomous learning by encouraging the conscious use of cognitive and metacognitive strategies.

5 DISCUSSION

The results confirm the effectiveness of the scaffolded suggestion system compared to direct correction in ELE writing. The experimental group showed significant improvements both in the immediate posttest and in retention after two weeks ($p < .05$), suggesting a more lasting consolidation of grammatical learning and greater activation of linguistic awareness. On the qualitative level, the higher prompts (L4–L5) fostered metacognitive processes such as self-regulation and critical evaluation. Although learners with lower proficiency (low B1) required more mediation to apply these suggestions effectively, the gradual scaffolding allowed adaptation to the learner’s profile. In addition, the system not only reduced linguistic errors but also promoted macrotextual revisions—such as paragraph reordering and connector reformulation—demonstrating its impact on overall coherence. A qualitative deep dive revealed three metacognitive strategies triggered by the hierarchical prompts: (1) Directed Substitution under L2–L3 prompts, where learners applied grammatical rules (e.g., gender–number agreement, tense selection) literally and efficiently; (2) Contextual Restructuring with L4 prompts, prompting learners to evaluate phrasing alternatives, adjust connectors, and reformulate sentences for coherence; and (3) Critical Evaluation under L5, encouraging justification of grammatical choices—even consulting external sources—demonstrating high self-regulation and reflection. Overall, this stepped-prompt approach, unlike traditional teacher feedback, fosters more active and lasting internalization of the target language norms.

The asynchronous interaction with ChatGPT enabled learners to manage their own revision pace, supporting autonomous reflective processes. Unlike time-constrained teacher feedback, generative AI provided a penalty-free space for practice, particularly valuable for students with language anxiety. Theoretically, the findings support the Zone of Proximal Development (ZPD) framework: progressively

scaffolded mediation activates higher-order cognitive functions. Unlike traditional automated correctors, the tiered model encourages active internalization of linguistic norms. This principle underpins the design of the proposed system, conceived as a progression of prompts from basic error identification to metalinguistic reflection. Finally, the study suggests practical applications: teachers can build banks of level-specific suggestions (e.g., L1: “Check subject–verb agreement”; L3: “Do you remember when to use the subjunctive?”) and integrate them as templates deployable through ChatGPT. This strategy transforms the chatbot into an intelligent tutor and promotes process-oriented teaching, in line with approaches such as Task-Based Language Learning.

6 CONCLUSIONS AND LIMITATIONS

In summary, this study demonstrates that scaffolded suggestions mediated by ChatGPT improve the quality of writing among learners of Spanish as a foreign language, with particularly positive effects on retention and the development of metacognitive strategies. It is concluded that this approach is especially suitable for B1-level learners or above, who already possess the necessary competencies to benefit from higher levels of cognitive guidance. In contrast, for beginner learners (A1–A2), it would be advisable to limit the complexity of prompts to levels L1–L3 and to supplement them with visual aids or more explicit examples. As with any study, there are some limitations. The influence of the learners’ native language on the interpretation of prompts was not examined in depth, which may be a critical factor in the system’s effectiveness (e.g., Chinese-speaking students tend to omit definite articles, which may require culturally adapted prompts). Additionally, the sample size and the short duration of the study (three writing tasks) limit the generalizability of the findings. To facilitate the practical adoption of this approach, instructors might develop repositories of L1–L5 prompts organized by grammatical objective and integrate them as templates within their LMS; automate the delivery of the appropriate prompt level following each written assignment submission; train in the selection of scaffolding levels according to learner profiles and the interpretation of ChatGPT interaction logs; establish formative review cycles (e.g., a monthly writing task) to iteratively adjust prompt complexity; and, for A1–A2 proficiency levels, combine L1–L3 prompts with visual supports or model examples to enhance comprehension. Future research could incorporate longitudinal analyses, include additional proficiency levels, and explore the integration of these systems into hybrid teaching platforms.

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